Math Chaos

Project Description:

I found my favorite game book (Math Games with Bad Drawings) at MIT Press in Kendall Square. This book introduces 75 math-based puzzle games that are intended to be played with pencil, paper, and dice. Our favorite is “SIM” which is a Ramsey theory (basically a mix of economics and probability theory named after Frank Ramsey) game where two players attempt play a multi-color connect-the-dots on the same hexagon shape while trying to force their opponent to make a triangle shape while defensively avoiding that outcome for themself.

User personas:

Addy Age: 21 | Computer Science student, Boston As a user, I want a quick and addictive strategy game to play during downtime without needing physical materials. I want to play against a bot and take my time between turns.

Bijesh and Cassandra Age: 29 and 30 | Software engineers, work from home As users, we want to play a game together online without downloading anything. We want to track our wins and losses over time and compete on a leaderboard.

Danielle the Planner Age: 34 | Marketing manager, two kids As a user, I want to learn graph theory and mathematical concepts by actively applying them in a game, in a way that reading a textbook would not.

User Stories:

User Story 1: Account Creation & Authentication As a player, I want to create a personal account so my stats and game history are saved and private. When I sign up I pick a username and password. The app remembers my session until I sign out, and my wins, losses, and streaks are all tied to my account.

User Story 2: Play vs AI As a solo player, I want to play SIM against an AI opponent so I can practice anytime without needing a friend. I can choose Easy or Hard difficulty. The AI responds immediately after my move and the game ends when either player completes a triangle in their own color.

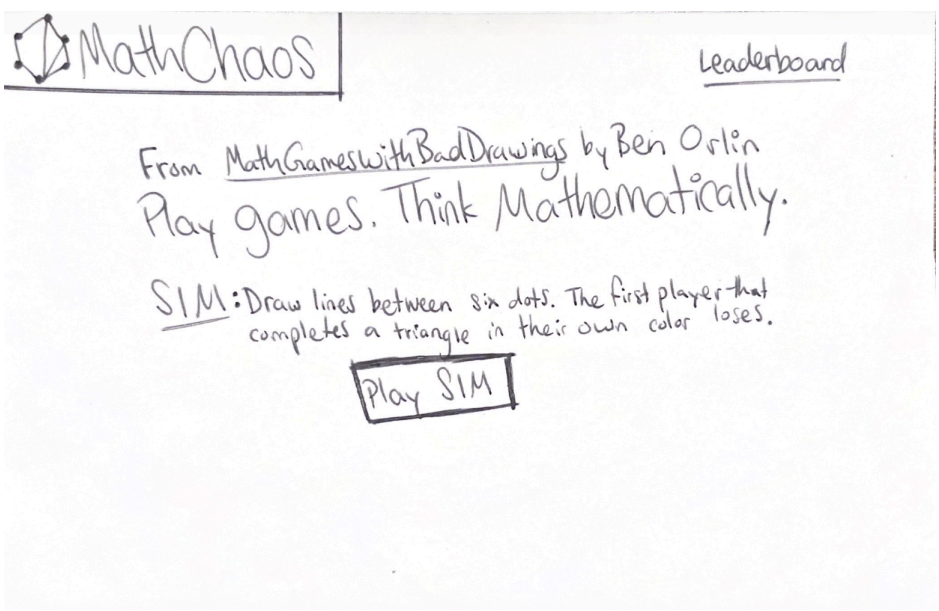
User Story 3: Play vs Player Online As a competitive player, I want to play SIM against a real opponent online. I create a room and share a room code with my friend. The game starts automatically when they join and our moves are synced in real time. If my opponent disconnects, I win by default.

User Story 4: Leaderboard As a competitive player, I want to see how I rank against other players. The leaderboard shows all players sorted by wins, win streak, or total games. I can see my own rank highlighted and load more players with pagination. I can also delete a player account from the leaderboard.

User Story 5: Game History & Stats As a player, I want my wins, losses, total games, and win streak tracked automatically after every game so I can see my progress over time. Stats update immediately after a game ends.

User Story 6: Learn While Playing As a curious player, I want to understand the math behind SIM after each game. The game over screen explains Ramsey Theory and why a draw is mathematically impossible in SIM — R(3,3) = 6 — with links to further reading.

Design Mockups:



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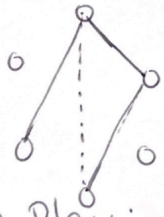
SIM

A classic game theory and strategy game. Try not to make a triangle within the hexagon.



How to Play:
1. Draw a line between dots

Account

username

game mode

AI | Human

difficulty

Easy | Hard

Play

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Your Turn

Games

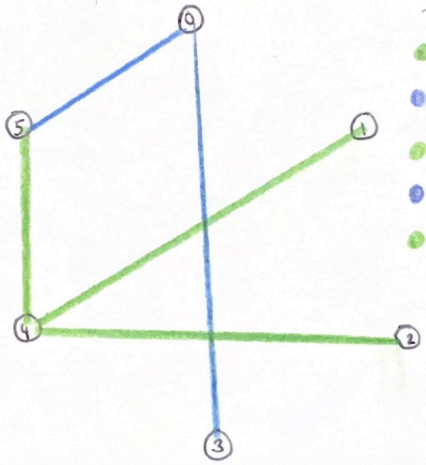
Move History

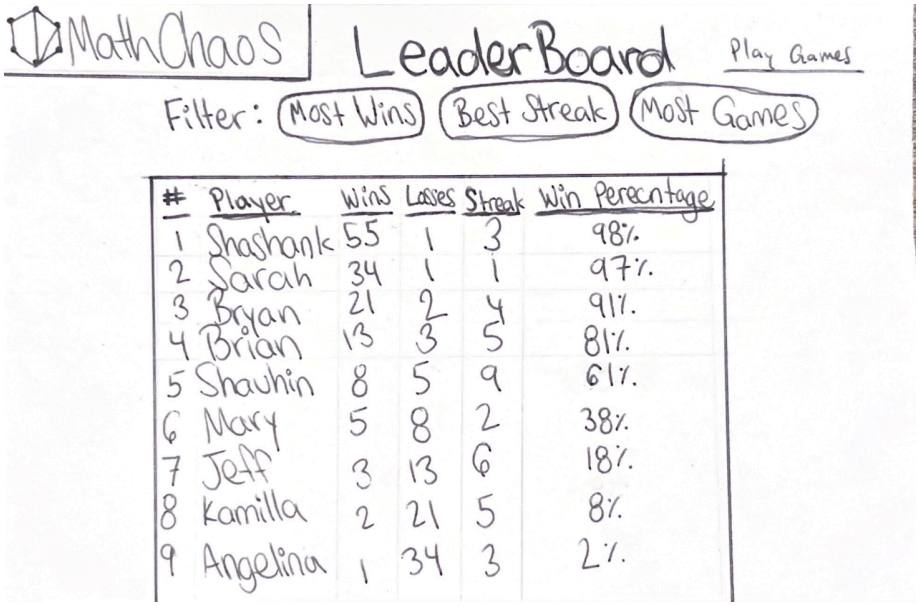
- 1-4
- 5-0
- 2-4
- 3-0
- 5-4

Players

Sarah (You)

AI Made





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LeaderBoard

Play Games

Filter: Most Wins | Best Streak | Most Games

#	Player	Wins	Losses	Streak	Win Percentage
1	Shashank	55	1	3	98%
2	Sarah	34	1	1	97%
3	Bryan	21	2	4	91%
4	Brian	13	3	5	81%
5	Shauhin	8	5	9	61%
6	Mary	5	8	2	38%
7	Jeff	3	13	6	18%
8	Kamilla	2	21	5	8%
9	Angelina	1	34	3	2%

Our design document on how to structure the database:

users collection
Stores authentication credentials. Separate from players to keep auth concerns isolated.

```
{
  username: string,        — unique, original casing stored
  password: string,        — bcrypt hashed
  createdAt: date
}
```

players collection
Stores game stats for each player. Created automatically at signup.

```
{
  username: string,        — matches users.username
  wins: number,
  losses: number,
  totalGames: number,
  winStreak: number,
  bestStreak: number,
  createdAt: date,
  updatedAt: date
}
```

games collection
Stores every game session — both finished and in progress.

```
{
  gameName: string,        — "sim" (supports future games)
  mode: string,            — "ai" | "multiplayer"
  players: [
    { username: string, color: "red" | "blue" }
  ],
  difficulty: string,      — "easy" | "hard" | null (multiplayer)
  moves: [
    { edge: [number, number], color: string, username: string, timestamp: date }
  ],
  currentTurn: string,     — "red" | "blue"
  winner: string,          — username of winner | null
  status: string,          — "waiting" | "active" | "finished"
  losingTriangle: [number, number, number] | null,
  roomCode: string,        — last 6 chars of _id, multiplayer only
  blueUsername: string,    — multiplayer joiner username
  durationSeconds: number,
  createdAt: date,
  updatedAt: date
}
```